

Overview of Seaborn

What is Seaborn?

Seaborn is a Python data visualization library built on top of Matplotlib. It provides a high-level interface for drawing attractive and informative statistical graphics. Seaborn integrates closely with pandas DataFrames, making it the go-to library for exploratory data analysis in data science and machine learning workflows.

How to Install:

```
pip install seaborn
```

Code Example:

```
import seaborn as sns
import matplotlib.pyplot as plt

# Load a built-in dataset
tips = sns.load_dataset("tips")

# Create a simple scatter plot
sns.scatterplot(data=tips, x="total_bill", y="tip", hue="sex")
plt.title("Tips Dataset - Scatter Plot")
plt.show()
```

Some commonly used methods:

Method	Description
scatterplot()	Draw a scatter plot with optional hue/size/style grouping.
lineplot()	Draw a line plot with optional confidence intervals.
barplot()	Show point estimates with error bars as bars.
histplot()	Plot univariate or bivariate histograms.
kdeplot()	Plot kernel density estimate curves.
boxplot()	Draw a box-and-whisker plot.
violinplot()	Combine a box plot with a kernel density estimate.
heatmap()	Plot rectangular data as a color-encoded matrix.
pairplot()	Plot pairwise relationships in a dataset.

Method	Description
jointplot()	Draw a bivariate plot with marginal univariate plots.
lmpplot()	Plot data and regression model fits across a grid.
catplot()	Figure-level interface for categorical plots.
relplot()	Figure-level interface for relational plots.
displot()	Figure-level interface for distribution plots.
FacetGrid()	Create a grid of plots based on data subsets.
set_theme()	Set the default aesthetic style for all plots.
set_palette()	Set the default color palette.

Some widely used Plot types:

1. Scatter Plot:

A scatter plot shows the relationship between two numeric variables. Seaborn's `scatterplot()` supports grouping by a third variable using hue, size, and style.

Code example:

```
import seaborn as sns
import matplotlib.pyplot as plt

tips = sns.load_dataset("tips")
sns.scatterplot(data=tips, x="total_bill", y="tip",
                hue="sex", size="size", style="time")
plt.title("Scatter Plot - Tips Dataset")
plt.show()
```

Formatting style methods for Scatter Plot:

Method	Definition	Code Example
hue	Groups data points by color.	<code>sns.scatterplot(..., hue="sex")</code>
size	Groups data points by marker size.	<code>sns.scatterplot(..., size="size")</code>
style	Groups data points by marker style.	<code>sns.scatterplot(..., style="time")</code>
palette	Sets the color palette for hue groups.	<code>sns.scatterplot(..., palette="Set2")</code>
alpha	Controls the transparency of points.	<code>sns.scatterplot(..., alpha=0.6)</code>
markers	Custom marker shapes per style group.	<code>sns.scatterplot(..., markers=True)</code>
edgecolor	Sets the edge color of the markers.	<code>sns.scatterplot(..., edgecolor="w")</code>
linewidth	Width of the marker edge.	<code>sns.scatterplot(..., linewidth=0.5)</code>

2. Line Plot:

A line plot is used to visualize trends over a continuous variable. Seaborn automatically computes and displays a confidence interval around the estimate when multiple y-values exist for each x.

Code example:

```
import seaborn as sns
import matplotlib.pyplot as plt

flights = sns.load_dataset("flights")
flights_pivot = flights.pivot(index="month", columns="year", values="passengers")
sns.lineplot(data=flights, x="year", y="passengers", hue="month")
plt.title("Passengers per Year")
plt.show()
```

Formatting style methods for Line Plot:

Method	Definition	Code Example
hue	Assigns a different line color per group.	<code>sns.lineplot(..., hue="month")</code>
style	Assigns a different line style per group.	<code>sns.lineplot(..., style="event")</code>
dashes	Custom dash pattern per style group.	<code>sns.lineplot(..., dashes=False)</code>
markers	Add markers at data points.	<code>sns.lineplot(..., markers=True)</code>
ci	Size of the confidence interval band (deprecated).	<code>sns.lineplot(..., ci=95)</code>
errorbar	Method for computing error bars.	<code>sns.lineplot(..., errorbar='sd')</code>
estimator	Function to aggregate multiple y values.	<code>sns.lineplot(..., estimator="mean")</code>
palette	Color palette for hue groups.	<code>sns.lineplot(..., palette="tab10")</code>

3. Bar Plot:

A bar plot shows point estimates (default: mean) and confidence intervals for a numeric variable across categories. It is useful for comparing group-level summaries.

Code example:

```
import seaborn as sns
import matplotlib.pyplot as plt

tips = sns.load_dataset("tips")
sns.barplot(data=tips, x="day", y="total_bill", hue="sex",
            palette="pastel", capsize=0.1)
plt.title("Average Total Bill by Day")
plt.show()
```

Formatting style methods for Bar Plot:

Method	Definition	Code Example
hue	Groups bars by color.	sns.barplot(..., hue="sex")
palette	Color palette for bars.	sns.barplot(..., palette="Set3")
capsize	Width of the error bar caps.	sns.barplot(..., capsize=0.1)
estimator	Statistical function for aggregation.	sns.barplot(..., estimator=np.median)
errorbar	Error-bar method ('ci', 'pi', 'se', 'sd').	sns.barplot(..., errorbar='sd')
order	Order of the categorical levels.	sns.barplot(..., order=['Sat', 'Sun'])
width	Width of each bar.	sns.barplot(..., width=0.5)
saturation	Proportion of saturation for bar colors.	sns.barplot(..., saturation=0.8)

4. Hist Plot:

histplot() displays the distribution of one or two variables by dividing data into bins. It can optionally overlay a KDE curve and supports bivariate histograms.

Code example:

```
import seaborn as sns
import matplotlib.pyplot as plt

penguins = sns.load_dataset("penguins")
sns.histplot(data=penguins, x="flipper_length_mm",
             hue="species", bins=30, kde=True)
plt.title("Flipper Length Distribution")
plt.show()
```

Formatting style methods for Hist Plot:

Method	Definition	Code Example
bins	Number of bins or bin edges.	<code>sns.histplot(..., bins=20)</code>
kde	Overlays a kernel density estimate curve.	<code>sns.histplot(..., kde=True)</code>
hue	Groups histogram by color.	<code>sns.histplot(..., hue="species")</code>
stat	Aggregate stat: count, frequency, density, etc.	<code>sns.histplot(..., stat='density')</code>
cumulative	Plots the cumulative distribution.	<code>sns.histplot(..., cumulative=True)</code>
fill	Fills the area under the histogram bars.	<code>sns.histplot(..., fill=True)</code>
element	Visual representation: bars, step, poly.	<code>sns.histplot(..., element='step')</code>
multiple	How to handle multiple distributions.	<code>sns.histplot(..., multiple='stack')</code>

5. KDE Plot:

A Kernel Density Estimate (KDE) plot shows a smooth estimate of the distribution of data. It can be used for univariate or bivariate distributions and is often layered on histograms.

Code example:

```
import seaborn as sns
import matplotlib.pyplot as plt

tips = sns.load_dataset("tips")
sns.kdeplot(data=tips, x="total_bill", hue="time",
            fill=True, alpha=0.5, palette="crest")
plt.title("KDE Plot - Total Bill by Time")
plt.show()
```

Formatting style methods for KDE Plot:

Method	Definition	Code Example
hue	Groups curves by color.	sns.kdeplot(..., hue="time")
fill	Fills the area under the KDE curve.	sns.kdeplot(..., fill=True)
bw_adjust	Bandwidth adjustment factor.	sns.kdeplot(..., bw_adjust=0.5)
cut	Factor that extends the range of the curve.	sns.kdeplot(..., cut=3)
cumulative	Plots the cumulative distribution function.	sns.kdeplot(..., cumulative=True)
multiple	How overlapping KDEs are drawn.	sns.kdeplot(..., multiple='stack')
levels	Number of contour levels for bivariate plots.	sns.kdeplot(x=x, y=y, levels=5)
alpha	Transparency of the filled area.	sns.kdeplot(..., alpha=0.4)

6. Box Plot:

A box plot displays the five-number summary (minimum, Q1, median, Q3, maximum) of a numeric variable across categories. It is effective for spotting outliers and comparing distributions.

Code example:

```
import seaborn as sns
import matplotlib.pyplot as plt

tips = sns.load_dataset("tips")
sns.boxplot(data=tips, x="day", y="total_bill",
            hue="smoker", palette="Set2")
plt.title("Total Bill Distribution by Day")
plt.show()
```

Formatting style methods for Box Plot:

Method	Definition	Code Example
hue	Groups boxes by color.	sns.boxplot(..., hue="smoker")
palette	Color palette for boxes.	sns.boxplot(..., palette="Set2")
width	Width of each box.	sns.boxplot(..., width=0.5)
fliersize	Size of the outlier markers.	sns.boxplot(..., fliersize=4)
linewidth	Width of lines forming the box.	sns.boxplot(..., linewidth=1.5)
notch	Draws a notch around the median.	sns.boxplot(..., notch=True)
order	Order of categorical levels.	sns.boxplot(..., order=["Fri","Sat"])
saturation	Saturation of box fill color.	sns.boxplot(..., saturation=0.75)

7. Violin Plot:

A violin plot combines a box plot with a KDE to show the distribution of data. The width of the violin at each point indicates the density of observations there.

Code example:

```
import seaborn as sns
import matplotlib.pyplot as plt

tips = sns.load_dataset("tips")
sns.violinplot(data=tips, x="day", y="total_bill",
               hue="sex", split=True, palette="muted")
plt.title("Violin Plot - Total Bill by Day")
plt.show()
```

Formatting style methods for Violin Plot:

Method	Definition	Code Example
split	Draws half-violin for each hue level (two levels).	sns.violinplot(..., split=True)
inner	Interior representation: box, quart, point, stick.	sns.violinplot(..., inner='quart')
bw	Bandwidth method or scale factor.	sns.violinplot(..., bw='scott')
cut	Distance past extreme data points.	sns.violinplot(..., cut=2)
scale	Scales violin width: area, count, width.	sns.violinplot(..., scale='count')
palette	Color palette for violins.	sns.violinplot(..., palette="Set1")
linewidth	Width of the violin body lines.	sns.violinplot(..., linewidth=1)
saturation	Saturation of violin fill colors.	sns.violinplot(..., saturation=0.8)

8. Heatmap:

A heatmap visualizes a matrix of values using color encoding. It is commonly used to display correlation matrices, confusion matrices, and pivot tables.

Code example:

```

import seaborn as sns
import matplotlib.pyplot as plt

flights = sns.load_dataset("flights")
pivot = flights.pivot(index="month", columns="year",
                       values="passengers")
sns.heatmap(pivot, annot=True, fmt="d", cmap="YlOrRd")
plt.title("Flights Heatmap")
plt.show()

```

Formatting style methods for Heatmap:

Method	Definition	Code Example
annot	Annotates each cell with its value.	<code>sns.heatmap(..., annot=True)</code>
fmt	Format string for cell annotations.	<code>sns.heatmap(..., fmt=".2f")</code>
cmap	Colormap for the heatmap.	<code>sns.heatmap(..., cmap="coolwarm")</code>
vmin/vmax	Anchors the colormap scale.	<code>sns.heatmap(..., vmin=0, vmax=100)</code>
linewidths	Width of lines that divide cells.	<code>sns.heatmap(..., linewidths=0.5)</code>
linecolor	Color of the dividing lines.	<code>sns.heatmap(..., linecolor='white')</code>
mask	Masks cells where mask is True.	<code>sns.heatmap(..., mask=mask_df)</code>
square	Forces equal-sized cells.	<code>sns.heatmap(..., square=True)</code>

9. Pair Plot:

pairplot() creates a grid of axes showing pairwise relationships in a dataset. Diagonal axes display univariate distributions of each variable, while off-diagonal axes show bivariate scatter plots.

Code example:

```
import seaborn as sns
import matplotlib.pyplot as plt

iris = sns.load_dataset("iris")
sns.pairplot(iris, hue="species", diag_kind="kde",
             plot_kws={"alpha": 0.6})
plt.suptitle("Pair Plot - Iris Dataset", y=1.02)
plt.show()
```

Formatting style methods for Pair Plot:

Method	Definition	Code Example
hue	Color-codes points by a categorical variable.	sns.pairplot(..., hue="species")
diag_kind	Type of plot on the diagonal: auto, hist, kde.	sns.pairplot(..., diag_kind='kde')
kind	Type of off-diagonal plot: scatter, kde, hist, reg.	sns.pairplot(..., kind='reg')
vars	Subset of variables to include.	sns.pairplot(..., vars=['a','b'])
x_vars/y_vars	Different variables for rows and columns.	sns.pairplot(..., x_vars=["a"])
plot_kws	Extra keyword arguments for off-diagonal plots.	sns.pairplot(..., plot_kws={'alpha':0.5})
diag_kws	Extra keyword arguments for diagonal plots.	sns.pairplot(..., diag_kws={'bins':20})
palette	Color palette for hue levels.	sns.pairplot(..., palette="husl")

10. Joint Plot:

jointplot() draws a bivariate plot with marginal univariate distributions on the axes. It shows both the joint and marginal distributions together in a single figure.

Code example:

```
import seaborn as sns
import matplotlib.pyplot as plt

tips = sns.load_dataset("tips")
sns.jointplot(data=tips, x="total_bill", y="tip",
              kind="reg", color="#4C72B0")
plt.suptitle("Joint Plot - Total Bill vs Tip", y=1.02)
plt.show()
```

Formatting style methods for Joint Plot:

Method	Definition	Code Example
kind	Type of plot: scatter, kde, hist, hex, reg, resid.	sns.jointplot(..., kind='kde')
color	Color for the plot elements.	sns.jointplot(..., color="g")
ratio	Size ratio between joint and marginal axes.	sns.jointplot(..., ratio=5)
space	Space between joint and marginal axes.	sns.jointplot(..., space=0.1)
marginal_kws	Keyword arguments for marginal plots.	sns.jointplot(..., marginal_kws={'bins':20})
joint_kws	Keyword arguments for the joint plot.	sns.jointplot(..., joint_kws={'alpha':0.5})
hue	Semantic grouping by color.	sns.jointplot(..., hue="sex")
xlim/ylim	Axis limits for the joint plot.	sns.jointplot(..., xlim=(0,60))

11. FacetGrid:

FacetGrid creates a grid of subplots based on the values of one or more categorical variables. It allows the same plot to be drawn for each subset of data, making comparison easy.

Code example:

```
import seaborn as sns
import matplotlib.pyplot as plt

tips = sns.load_dataset("tips")
g = sns.FacetGrid(tips, col="time", row="sex", hue="smoker",
                  margin_titles=True)
g.map(sns.scatterplot, "total_bill", "tip", alpha=0.6)
g.add_legend()
plt.show()
```

Formatting style methods for FacetGrid:

Method	Definition	Code Example
col / row	Variables that define columns and rows.	FacetGrid(..., col="time")
hue	Variable for color grouping.	FacetGrid(..., hue="smoker")
col_wrap	Wraps columns into multiple rows.	FacetGrid(..., col_wrap=3)
height	Height of each facet in inches.	FacetGrid(..., height=4)
aspect	Aspect ratio of each facet.	FacetGrid(..., aspect=1.2)
margin_titles	Places row titles on the right margin.	FacetGrid(..., margin_titles=True)
sharex/sharey	Shares axes across facets.	FacetGrid(..., sharex=False)
palette	Color palette for hue groups.	FacetGrid(..., palette="Set1")

Themes and Colour Palettes:

Seaborn provides built-in themes and colour palettes to control the overall visual style of plots. Themes can be set globally or applied temporarily using a context manager.

Code example:

```

import seaborn as sns
import matplotlib.pyplot as plt

# Set global theme
sns.set_theme(style="whitegrid", palette="pastel", font_scale=1.2)

# Temporarily override style
with sns.axes_style("darkgrid"):
    sns.lineplot(x=[1,2,3], y=[1,4,9])
    plt.show()

# Inspect available palettes
sns.color_palette("husl", 8)

```

Available Styles and Palettes:

Category	Options
Styles	darkgrid, whitegrid, dark, white, ticks
Contexts	paper, notebook (default), talk, poster
Palettes	deep, muted, bright, pastel, dark, colorblind, husl, Set1–3, tab10, viridis, ...
set_theme()	Combines style + palette + font settings in one call
set_context()	Scales line widths, font sizes for different output media
despine()	Removes top and right axis spines from a plot

Figure-level vs Axes-level Functions:

Seaborn distinguishes between figure-level and axes-level functions. Figure-level functions (`relplot`, `displot`, `catplot`) manage their own figure and return a `FacetGrid`. Axes-level functions (`scatterplot`, `histplot`, `boxplot`) draw onto an existing Matplotlib axes.

Category	Functions	Returns
Figure-level	<code>relplot()</code> , <code>displot()</code> , <code>catplot()</code>	<code>FacetGrid</code>
Relational	<code>scatterplot()</code> , <code>lineplot()</code>	<code>Axes</code>
Distributional	<code>histplot()</code> , <code>kdeplot()</code> , <code>ecdfplot()</code> , <code>rugplot()</code>	<code>Axes</code>
Categorical	<code>stripplot()</code> , <code>swarmplot()</code> , <code>boxplot()</code> , <code>violinplot()</code> , <code>boxenplot()</code> , <code>pointplot()</code> , <code>barplot()</code> , <code>countplot()</code>	<code>Axes</code>
Regression	<code>lmpplot()</code> , <code>regplot()</code> , <code>residplot()</code>	<code>FacetGrid</code> / <code>Axes</code>
Matrix	<code>heatmap()</code> , <code>clustermap()</code>	<code>Axes</code> / <code>ClusterGrid</code>
Multi-plot grids	<code>FacetGrid()</code> , <code>PairGrid()</code> , <code>JointGrid()</code>	Grid object

Notes on Seaborn Usage:

1. Tight integration with pandas:

Seaborn functions accept a `data=` parameter with a `DataFrame`, along with column names for `x`, `y`, `hue`, `size`, and `style`. This keeps plotting code clean and readable.

2. Matplotlib compatibility:

Because Seaborn is built on Matplotlib, all plots can be further customised using standard `plt` or `ax` methods (e.g., `ax.set_title()`, `ax.set_xlim()`).

3. Built-in datasets:

Seaborn ships with several sample datasets (`tips`, `iris`, `flights`, `penguins`, `titanic`, etc.) loadable via `sns.load_dataset('name')`, ideal for practice.

4. Statistical estimation:

Many Seaborn functions automatically compute and display statistical aggregates (mean, median) and confidence intervals without extra code.